

Veronica Boyce

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EDUCATION

Stanford University

Doctor of Philosophy in Psychology

Dissertation: *Constraints on the emergence and interpretability of linguistic conventions*

Massachusetts Institute of Technology

Bachelor of Science in Brain & Cognitive Science

Stanford, CA

September 2025

Cambridge, MA

June 2019

EMPLOYMENT

MIT, Computational Psycholinguistics Lab

NSF SBE Postdoctoral Research Fellow

Stanford University, Language and Cognition Lab

Postdoctoral Researcher

MIT, Computational Psycholinguistics Lab

Research Support Associate

Cambridge, MA

Jan 2026 – present

Stanford, CA

Sep 2025 – Jan 2026

Cambridge, MA

June 2019 – July 2020

PUBLICATIONS

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- M. C. Frank, V. A. Marchman, C. A. Bergey, **V. Boyce**, M. Braginsky, G. Kachergis, J. Manke-witz, S. C. Meylan, B. Prystawski, N. Ram, R. Z. Sparks, A. Steffan, A. W. M. Tan, and M. Zettersten (2026). “Continuous developmental changes in word recognition support language learning across early childhood”. *eLife*. elifesciences.org/articles/109636.
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2024). “Interaction Structure Constrains the Emergence of Conventions in Group Communication”. *PNAS*. doi.org/10.1073/pnas.2403888121.
- V. Boyce**, B. Prystawski, A. B. Abutto, E. M. Chen, Z. Chen, H. Chiu, I. Ergin, A. Gupta, C. Hu, B. Kemmann, N. Klevak, V. Y. Q. Lua, M. Mazzaferro, K. Mon, D. Ogunbamowo, A. Pereira, J. Troutman, S. Tung, R. Uricher, and M. C. Frank (2024). “Estimating the replicability of psychology experiments after an initial failure to replicate”. *Collabra: Psychology*. doi.org/10.1525/collabra.125685.
- A. Chuey, **V. Boyce**, C. Anjie, and M. C. Frank (2024). “Conducting Developmental Research Online vs. in-Person: A Meta-Analysis”. *Open Mind*. doi.org/10.1162/opmi_a_00147.
- M. Soderstrom, J. Rocha-Hidalgo, L. E. Muñoz, A. Bochynska, J. F. Werker, B. Skarabela, A. Seidl, Y. Ryjova, J. L. Rennels, C. E. Potter, M. Paulus, M. Ota, N. M. Olesen, K. M. Nave, J. Mayor, A. Martin, L. C. Machon, C. Lew-Williams, E.-S. Ko, H. Kim, N. Kartushina, M. Kammermeier, A. Jessop, J. F. Hay, N. Havron, E. E. Hannon, J. K. Hamlin, N. Gonzalez-Gomez, A. Gampe, T. Fritzsche, M. C. Frank, S. Durrant, C. Davies, C. Cashon, K. Byers-Heinlein, **V. Boyce**, A. K. Black, C. Bergmann, L. Anderson, M. K. Alshakhori, A. H. Al-Hoorie, and A. S. M. Tsui (2024). “Testing the relationship between preferences for infant-directed speech and vocabulary development: A multi-lab study”. *Journal of Child Language*. doi.org/10.1017/S0305000924000254.

- V. Boyce** and R. P. Levy (2023). “A-Maze of Natural Stories: Comprehension and Surprisal in the Maze Task”. *Glossa Psycholinguistics*. escholarship.org/uc/item/6vh9d8zm.
- V. Boyce**, M. Mathur, and M. C. Frank (2023). “Eleven Years of Student Replication Projects Provide Evidence on the Correlates of Replicability in Psychology”. *Royal Society Open Science*. doi.org/10.1098/rsos.231240.
- M. Zettersten, D. Yurovsky, T. L. Xu, S. Uner, A. S. M. Tsui, R. M. Schneider, A. N. Saleh, S. C. Meylan, V. A. Marchman, J. Mankewitz, K. MacDonald, B. Long, M. Lewis, G. Kachergis, K. Handa, B. deMayo, A. Carstensen, M. Braginsky, **V. Boyce**, N. S. Bhatt, C. A. Bergey, and M. C. Frank (2023). “Peekbank: An Open, Large-Scale Repository for Developmental Eye-Tracking Data of Children’s Word Recognition”. *Behavior Research Methods*. doi.org/10.3758/s13428-022-01906-4.
- V. Boyce**, R. Futrell, and R. P. Levy (2020). “Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty”. *Journal of Memory and Language*. doi.org/10.1016/j.jml.2019.104082.

CONFERENCE PROCEEDINGS

- V. Boyce**, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). “Idiosyncratic but not opaque: Linguistic conventions formed in reference games are interpretable by naïve humans and vision–language models”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, preprint at osf.io/7d6sv_v1.
- V. Boyce**, R. Z. Sparks, Y. Mofor, and M. C. Frank (2025). “Preschoolers can form conventional pacts with each other to communicate about novel referents”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, osf.io/dpgfw_v1.
- A. W. M. Tan, B. Prystawski, **V. Boyce**, and M. C. Frank (2025). “Context informs pragmatic interpretation in vision-language models”. *CogInterp workshop NeurIPS 2025*. Poster, preprint at arxiv.org/abs/2511.03908.
- V. Boyce** and M. C. Frank (2023). “Communicative Reduction in Referring Expressions within a Multi-Player Negotiation Game”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/8dq8c2s6.
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2022). “Two’s Company but Six Is a Crowd: Emergence of Conventions in Multiparty Communication Games”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Talk, osf.io/preprints/psyarxiv/xd7ep.
- J. Mankewitz, **V. Boyce**, B. Waldon, G. Loukatou, D. Yu, J. Mu, N. D. Goodman, and M. C. Frank (2021). “Multi-Party Referential Communication in Complex Strategic Games”. *Meaning in Context: Pragmatic Communication in Humans and Machines workshop. NeurIPS*. PsyArXiv. Talk, osf.io/preprints/psyarxiv/tfb3d.
- M. Zettersten, C. A. Bergey, N. S. Bhatt, **V. Boyce**, M. Braginsky, A. Carstensen, B. deMayo, G. Kachergis, M. Lewis, B. Long, K. MacDonald, J. Mankewitz, S. C. Meylan, A. N. Saleh, R. M. Schneider, A. S. M. Tsui, S. Uner, T. L. Xu, D. Yurovsky, and M. C. Frank (2021). “Peekbank: Exploring Children’s Word Recognition through an Open, Large-Scale Repository for Developmental Eye-Tracking Data”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/9j05h4n1.
- V. Boyce**, A. Aravind, and M. Hackl (2017). “Lexical and Syntactic Effects on Auxiliary Selection: Evidence from Child French”. *Boston University Conference on Language Development*. Talk, lingref.com/bucl41/BUCLD41-08.pdf.

PREPRINTS (UNDER REVIEW)

- V. Boyce**, V. A. Marchman, H. A. Baumgartner, C. A. Bergey, M. Braginsky, F. Giovannetti, G. Kachergis, J. Mankewitz, S. C. Meylan, B. Prystawski, R. Z. Sparks, A. Steffan, A. W. M. Tan, M. C. Frank, et al. (2026). “Data-driven recommendations for increasing reliability and validity in measures of infant word recognition”. Preprint, under review, osf.io/preprints/psyarxiv/28gq4_v2.
- V. Boyce**, R. Z. Sparks, and M. C. Frank (2026). “Preschoolers can coordinate with each other to communicate about novel referents”. Preprint at osf.io/preprints/psyarxiv/t76a5_v2.
- A. N. Meltzoff, S. Farwaha, V. Slaughter, E. A. Simpson, R. Brooks, M. Nielsen, H. A. Baumgartner, L. Bayet, S. Benavides-Varela, **V. Boyce**, K. Byers-Heinlein, J. Dönicke, B. P. Flaherty, S. H. Forbes, M. C. Frank, et al. (2026). “ManyBabies 6: A Multi-Lab Investigation of Neonatal and Early Imitation”. Stage 1 Registered Report, under review, pre-print, osf.io/preprints/psyarxiv/fe7pj_v1.
- V. Boyce**, A. Acharya, and M. C. Frank (2025). “Use of communication channels in 2x2 strategic games”. White paper, osf.io/preprints/psyarxiv/y6ahw_v1.
- L. Röseler, L. Wallrich, H. Hartmann, L. Altegoer, **V. Boyce**, S. M. Field, J. Goltermann, J. Hüffmeier, C. R. Pennington, M.-M. Pittelkow, P. Silverstein, D. van Ravenzwaaij, and F. Azevedo (2025). “Handbook on Conducting Reproduction and Replication Studies”. Preprint, under revision following MetaROR peer review, doi.org/10.5281/zenodo.16990114.

CONFERENCE PRESENTATIONS

- V. Boyce**, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, B. Prystawski, M. C. Frank, and R. D. Hawkins (2026). “Refbank: an open repository of iterated reference games”. The 39th Annual Conference on Human Sentence Processing. Poster.
- V. Boyce** and R. P. Levy (2025). “Linguistic, but not visual, discourse context modulates syntactic parsing in the Maze”. The 38th Annual Conference on Human Sentence Processing. Poster.
- V. Boyce**, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). “Context matters: History and informativity affect understanding of opaque references”. The 38th Annual Conference on Human Sentence Processing. Poster.
- J. Vigly, **V. Boyce**, M. Socolof, J. Michaelov, T. von der Malsburg, and R. P. Levy (2025). “Implicit gender bias in linguistic descriptions for expected events: The case of the 2024 United States presidential election”. The 38th Annual Conference on Human Sentence Processing. Poster.
- V. Boyce**, I. Chen, R. Z. Sparks, M. Perez, and M. C. Frank (2024). “4-5 year old children can successfully communicate using ad-hoc referential expressions”. CAMP 6. Talk.
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). “Thick Feedback Facilitates Referential Coordination in Larger Groups”. CAMP 5. Talk.
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). “Thick Feedback Facilitates Referential Coordination in Larger Groups”. HSP. Poster.
- T. Poppels, **V. Boyce**, C. Ajunwa, T. von der Malsburg, and R. P. Levy (2021). “Bias against ‘she’ Pronouns Can Be Rapidly Overcome by Changing Event Expectations”. CUNY Human Sentence Processing Conference. Short talk.
- V. Boyce** and R. P. Levy (2020). “A-Maze of Natural Stories: Texts Are Comprehensible Using the Maze Task”. Architectures and Mechanisms for Language Processing. Talk.
- V. Boyce**, R. Futrell, and R. P. Levy (2019). “Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty”. CUNY Human Sentence Processing Conference. Poster.

- V. **Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). “Female Gender Is Consistently Under-Expressed in Pronoun Production and under-Inferred in Comprehension”. Linguistic Society of American Annual Meeting. Poster.
- V. **Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). “Remember ‘Him’, Forget ‘Her’: Gender Bias in the Comprehension of Pronominal Referents.” CUNY Human Sentence Processing Conference. Poster.
- V. **Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2018). “Implicit Gender Biases in the Production and Comprehension of Pronominal References”. CUNY Human Sentence Processing Conference. Poster.

GRANTS & FELLOWSHIPS

NSF SBE Postdoctoral Research Fellowship	Jan 2026-Jan 2028
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AWARDS

Open Science Innovator Prize, <i>Stanford Center for Open and Reproducible Science</i>	2025
The Albert H. and Barbara R. Hastorf Prize for Teaching, <i>Stanford Department of Psychology</i>	2025

TEACHING

Stanford, Department of Psychology	Stanford, CA
TA for PSYCH 125: Research Methods in Psychology	Jan 2024 – March 2024
<i>First offering of course; designed and led course replication projects with bi-weekly milestones</i>	
TA for PSYCH 251: Experimental Methods	Sep 2023 – Dec 2023
<i>Designed and supervised re-replication option for course project</i>	
<i>Led collective writing of joint paper reporting findings across projects (see Publications)</i>	
TA for PSYCH 251: Experimental Methods	Sep 2022 – Dec 2022
Guest Lecturer, PSYCH 251: Experimental Methods	2023, 2025
<i>Topic: meta-analysis and meta-science</i>	
TA for PSYCH 45: Introduction to Learning and Memory	April 2023 – June 2023
TA for PSYCH 220B: Probabilistic Models of Cognition: Language	Sep 2021 – Dec 2021
MIT, Department of Brain & Cognitive Science	Cambridge, MA
TA for Computational Psycholinguistics	February 2019 – May 2019

MENTORING

Benji Fernandez (Stanford, undergraduate honors thesis, co-advisor; thesis in progress)	2025–present
<i>Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints</i>	
Fiona Feng (Queens College undergraduate research assistant)	2025-2026
<i>Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints</i>	
Ryder Fried (Stanford undergraduate research assistant)	2025
<i>Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints</i>	

Spurti Nimbali (Stanford undergraduate research assistant)	2025
Zihan Wang (Stanford undergraduate research assistant)	2025
Yannick Mofor (Stanford undergraduate research assistant)	2024
<i>Co-author, CogSci proceedings, see Conference Proceedings</i>	
Rebecca Pizzitola (Stanford undergrad)	2024
Ilaria Chen (Stanford undergraduate research assistant)	2023
<i>Co-author, talk at CAMP, see Conference Presentations</i>	
Malia Perez (Stanford undergraduate research assistant)	2023
<i>Co-author, talk at CAMP, see Conference Presentations</i>	
Brittany Linus (Stanford undergraduate research assistant)	2022
Chelsea Ajunwa (MIT undergraduate research assistant)	2020
<i>Co-author, short talk at HSP, see Conference Presentations</i>	
Arun Wongprommoon (MIT undergraduate research assistant)	2019

AD HOC REVIEWING

Journals: Replication Research (2026); Journal of Personality and Social Psychology: Attitudes and Social Cognition (2026); Infant Behavior and Development; Open Mind (2025); Language, Cognition, and Neuroscience (2024); Royal Society Open Science (2024); Journal of Experimental Psychology: Learning, Memory, and Cognition (2023); Journal of Memory and Language (2023); Glossa Psycholinguistics (2021)

Co-reviewed (with Michael C. Frank): Psychological Science (2025), Perspectives on Psychological Science (2025), Nature Human Behavior (2024), Glossa Psycholinguistics (2023)

Conferences: CogSci (2021–2026), HSP (2026), California Meeting of Psycholinguists (2023), Society for Computation in Linguistics (2020–2026), Association for Computational Linguistics (2019)